

Division Detective Case Files

Unit 1: Number System Launch • Lesson 1-4 • EduWonderLab

UNIT 1 • LESSON 1-4 • TEACHER NOTES

Standard 6.NS.2

FORMAT True/false task-card investigation

TIME 25–30 min

GROUPING Pairs or small group

TOPIC Divide Multi-Digit Numbers

STANDARD 6.NS.2

PREP LEVEL Light — print and cut case-file cards

Learning goal *I can divide multi-digit numbers and check my quotient with multiplication.*

Teacher setup

- Print one case-file set per group. Each case shows a division and the detective's claimed quotient.
- Students solve, then decide whether the claim is true or false.
- Every case must be checked with multiplication before the verdict.

Materials

Case-file task cards, Scratch paper, Recording sheet, Pencil.

Differentiation & language support

- Graph paper helps keep digits lined up in long division (SPED).
- Sentence stem: "I checked my quotient by ___, so the claim is ___."
- ESOL: pre-teach the words dividend, divisor, quotient with a labeled example.

Key vocabulary (English / Español):

Term	Español	What it means
dividend	dividendo	the number being divided
divisor	divisor	the number you divide by
quotient	cociente	the answer to a division problem

Quick teacher check: *Collect verdicts for Cases 2 and 5 (the false ones).*

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Name: _____

Date: _____

Class: _____

Your goal: *I can divide multi-digit numbers and check my quotient with multiplication.*

What you need

Case-file cards, Scratch paper, Recording sheet.

How to play

1. Pick a case file. Solve the division on scratch paper.
2. Multiply your quotient by the divisor to check.
3. Decide if the detective's claim is True or False. If false, write the correct quotient.
4. Record your verdict and move to the next case.

Case Files

Solve each case, check with multiplication, then mark the detective's claim True or False.

1. Case 1: $1,344 \div 6$. Claim: 224. True or false?	2. Case 2: $2,952 \div 8$. Claim: 359. True or false?
3. Case 3: $4,725 \div 25$. Claim: 189. True or false?	4. Case 4: $6,016 \div 16$. Claim: 376. True or false?
5. Case 5: $9,184 \div 28$. Claim: 318. True or false?	6. Case 6: $3,510 \div 15$. Claim: 234. True or false?
7. Case 7: $7,488 \div 24$. Claim: 312. True or false?	8. Case 8: $5,733 \div 7$. Claim: 819. True or false?

Record your work (show each division and check):

Explain your thinking

Pick a false case. Explain the mistake the detective probably made.

Sentence frame: *The claim is ___ because ___ $\times$ ___ = ___.*

★ **Challenge:** Write your own case file with a false claim. Trade with a partner and catch each other's error.

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Case 1: $1,344 \div 6$. Claim: 224. True or false? Answer: True — $6 \times 224 = 1,344$	2. Case 2: $2,952 \div 8$. Claim: 359. True or false? Answer: False — correct quotient is 369
3. Case 3: $4,725 \div 25$. Claim: 189. True or false? Answer: True — $25 \times 189 = 4,725$	4. Case 4: $6,016 \div 16$. Claim: 376. True or false? Answer: True — $16 \times 376 = 6,016$
5. Case 5: $9,184 \div 28$. Claim: 318. True or false? Answer: False — correct quotient is 328	6. Case 6: $3,510 \div 15$. Claim: 234. True or false? Answer: True — $15 \times 234 = 3,510$
7. Case 7: $7,488 \div 24$. Claim: 312. True or false? Answer: True — $24 \times 312 = 7,488$	8. Case 8: $5,733 \div 7$. Claim: 819. True or false? Answer: True — $7 \times 819 = 5,733$

Teaching notes & look-fors

- False cases: Case 2 (369) and Case 5 (328).
- Multiplication check is the proof — require it on the recording sheet.